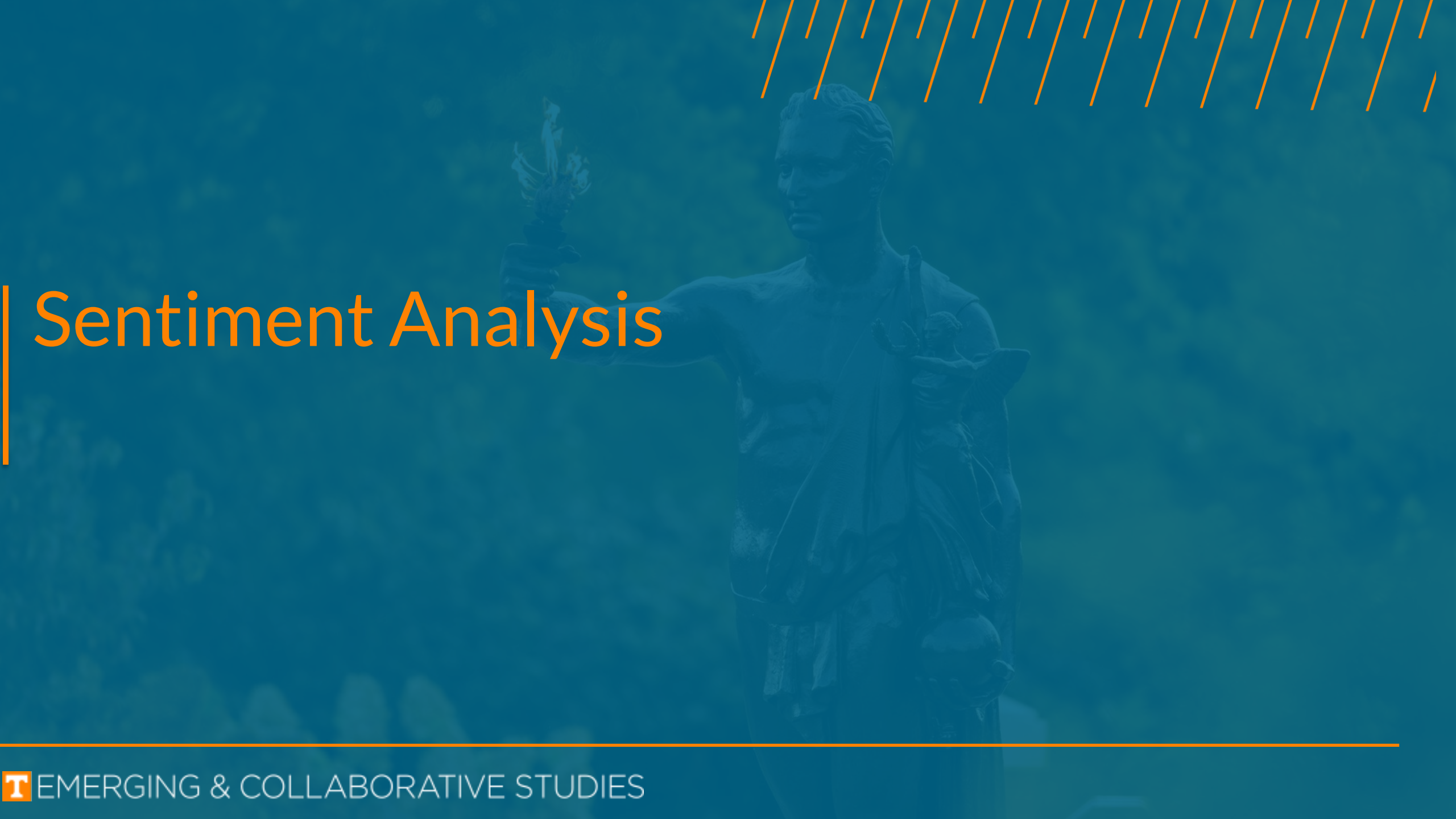

AI-102 Natural Language Processing (NLP)

Dr. Jimmy Ardiansyah



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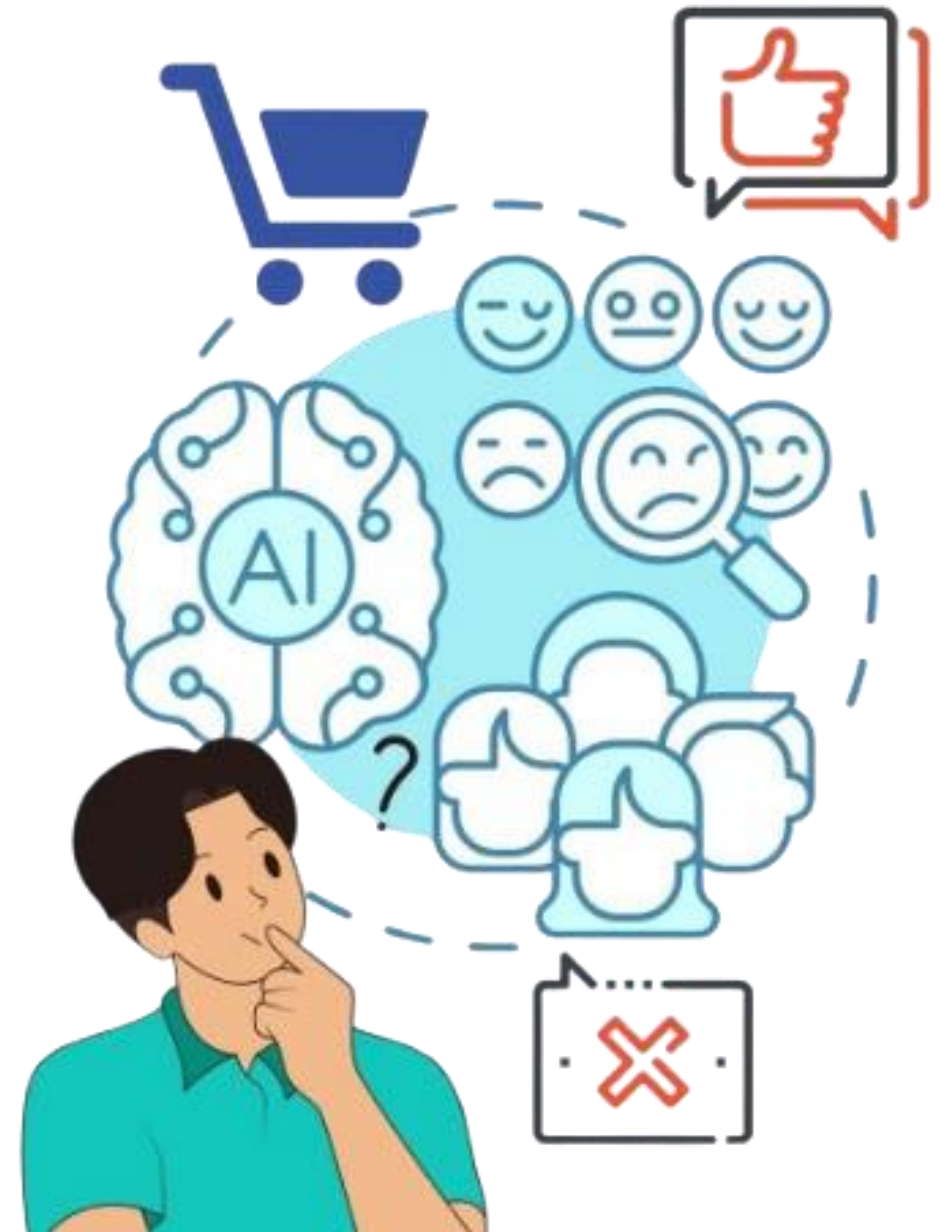




Sentiment Analysis

Rules-Based Approaches

Rule-based sentiment analysis uses hand-crafted rules plus sentiment lexicons (word lists with polarity and intensity) to assign an overall sentiment to text. It is simple, transparent, and easy to customize for a domain or language.

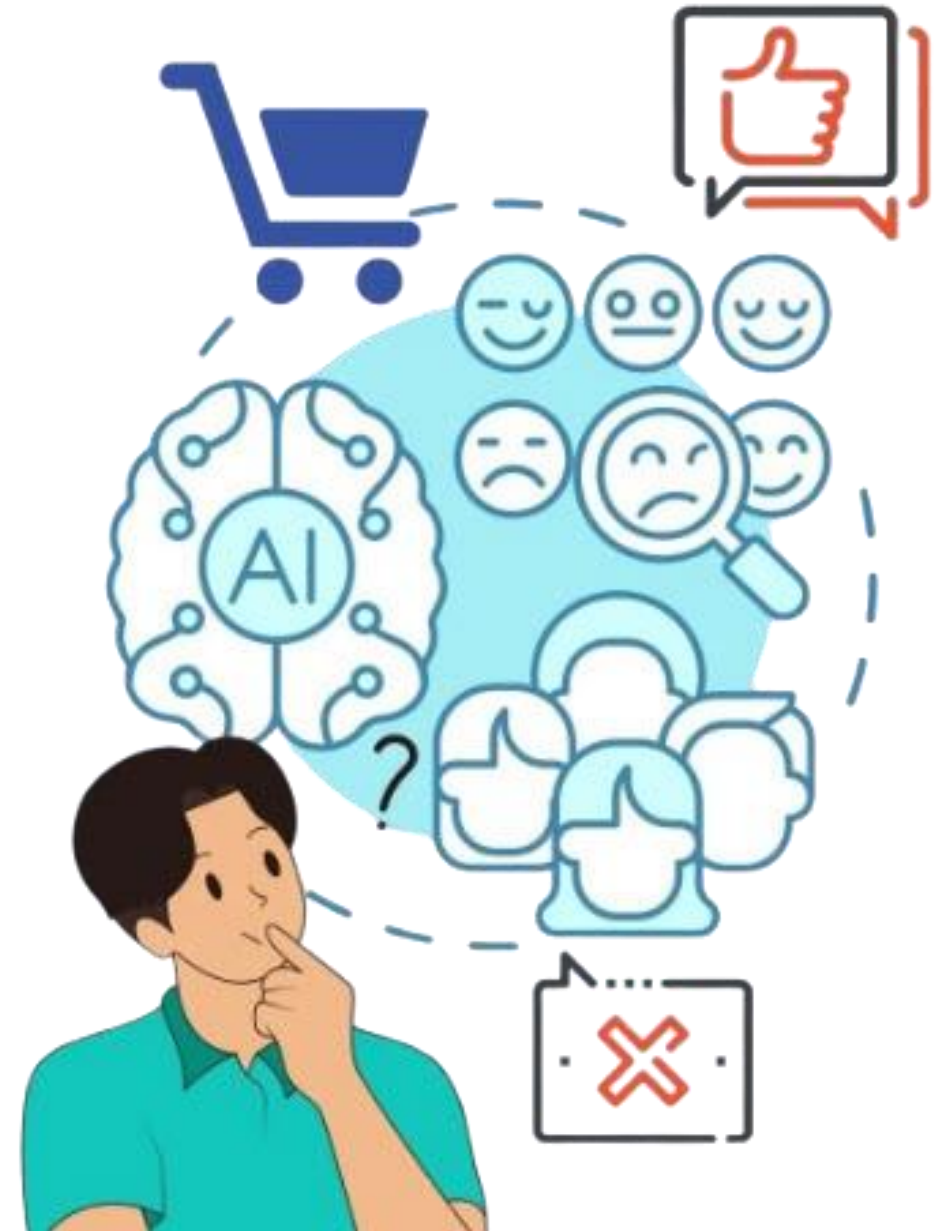


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Rule-Based Approaches Pipeline

- **Tokenization:** split into words or tokens.
- **Normalization:** lowercase, remove punctuation, standardize forms.
- **Lexicon lookup:** assign scores from resources like AFINN, SentiWordNet, NRC.
- **Rule application:** combine token scores and apply modifiers to decide overall sentiment.



Rules-Based Approaches

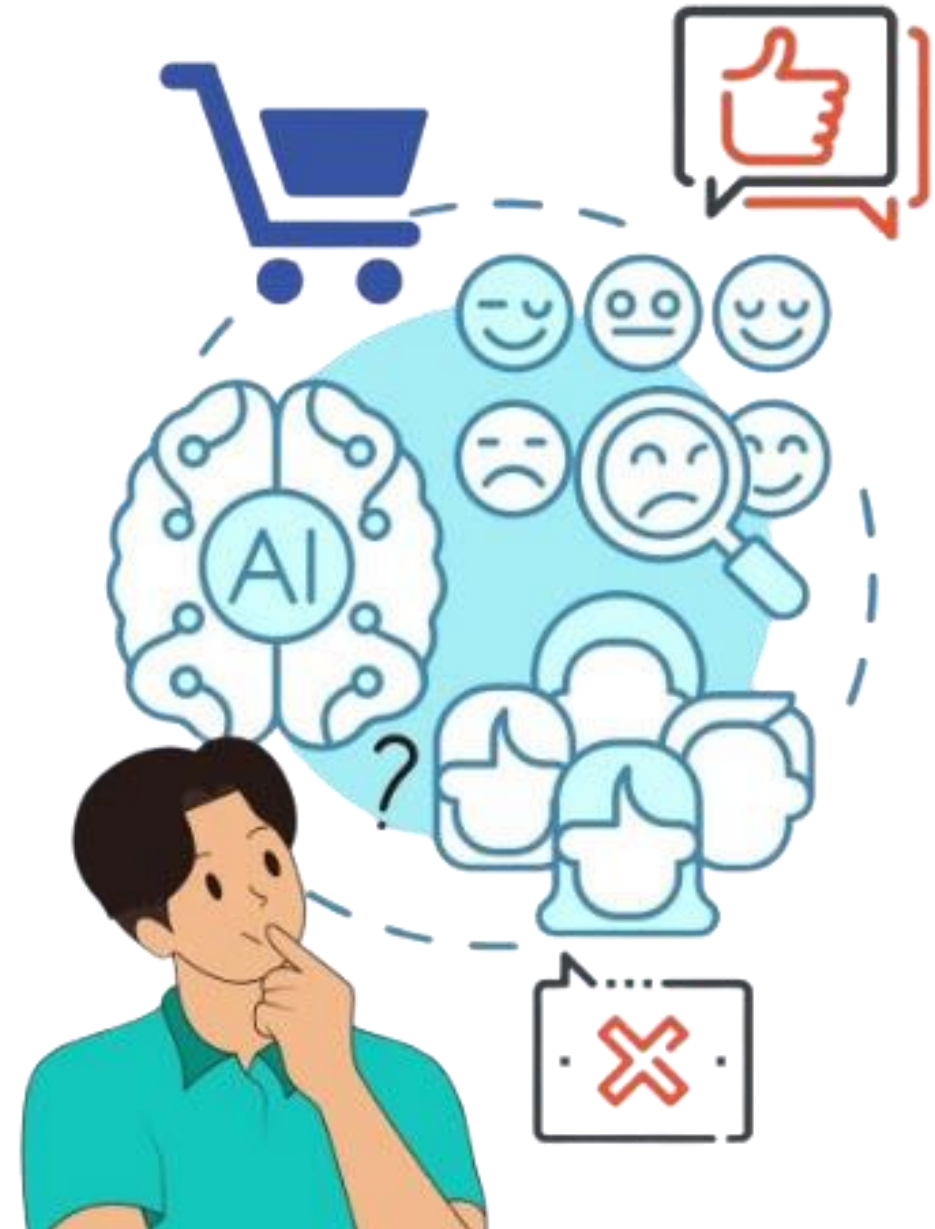
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Key Lexicons

- **AFINN:** integer scores -5 to +5 for words.
- **SentiWordNet:** scores at synset level for finer nuance.
- **NRC Emotion Lexicon:** maps words to emotions (joy, anger, etc.) and polarity.



Advantages and Limitation Rules-Based Approaches



- **Interpretable:** rules are explicit and reviewable.
- **Simple:** no heavy compute or large training sets needed.
- **Customizable:** add domain terms and patterns.



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- **Coverage:** slang and new expressions are missed.
- **Context:** struggles with sarcasm and irony.
- **Maintenance:** rules and lexicons need updates.

Practical Applications - Rules-Based Approaches

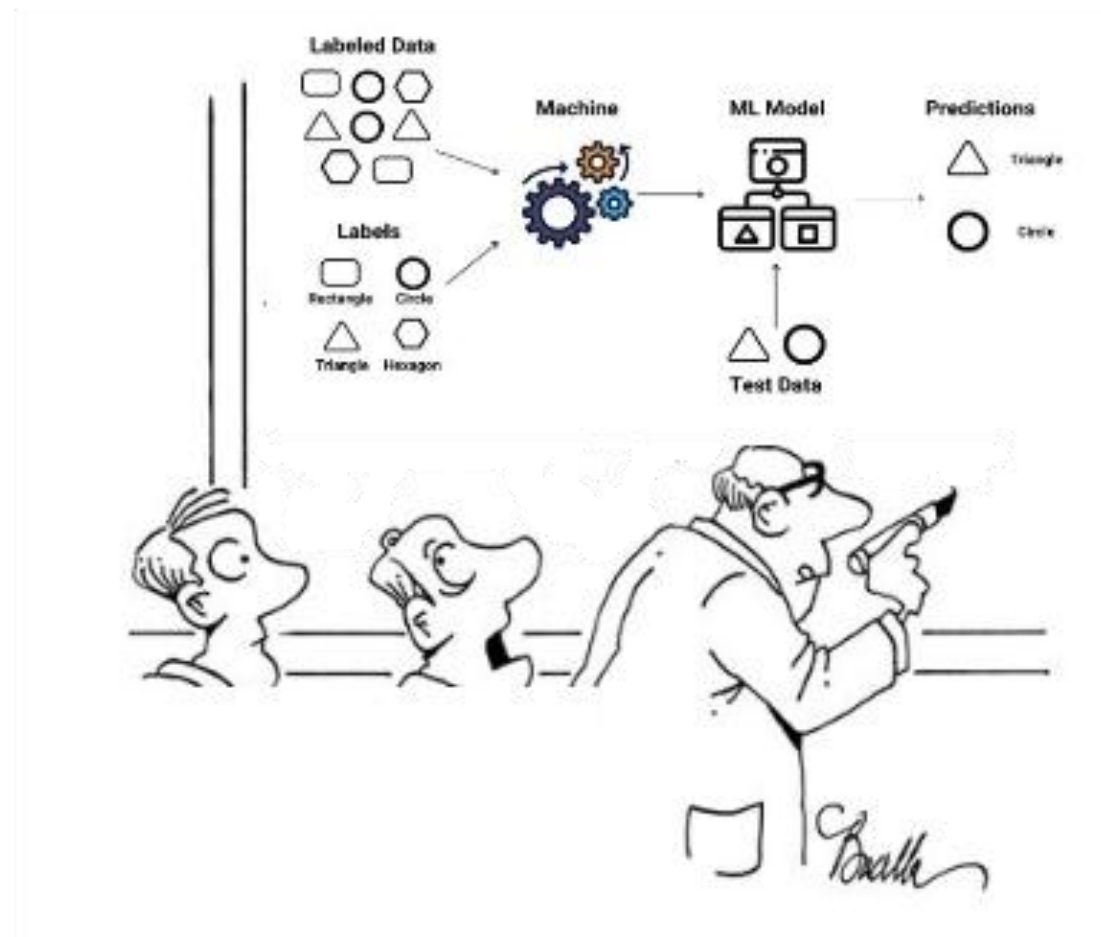


Sentiment analysis is a powerful tool with a wide range of practical applications across various industries. By understanding the sentiment behind textual data, companies and organizations can gain valuable insights that drive decision-making and strategy formulation. Here are some key applications:

- Customer Feedback Analysis
- Social Media Monitoring
- Market Research
- Brand Management
- Financial Market Analysis
- Healthcare and Public Health

Machine Learning Approaches

- Trains models on labeled data to learn sentiment patterns automatically.
- Uses algorithms like SVMs, neural networks, and ensemble methods.
- Predicts sentiment for new, unseen text with high accuracy.
- Outperforms rule-based systems that rely on fixed linguistic rules.
- Captures complex relationships, nuance, slang, and idiomatic expressions.
- More robust and adaptable for real-world sentiment analysis tasks.



Machine Learning Approaches Pipeline



Feature Extraction (Key Methods)

Bag of Words

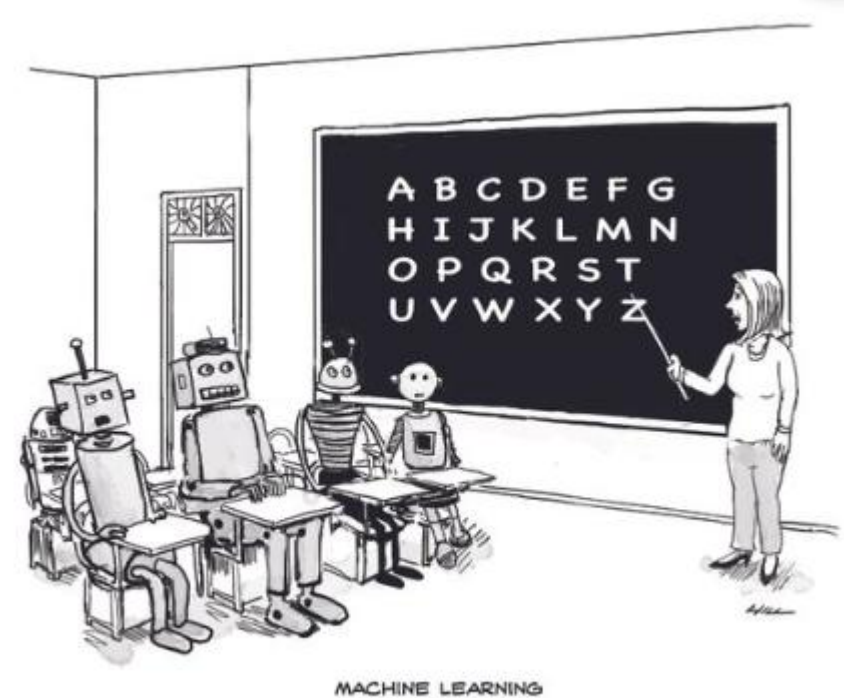
- Counts how many times each word appears.
- Simple, ignores word order.

TF-IDF

- Weighs words
- Highlights important, informative words.

Word Embeddings

- Dense vectors capturing semantic meaning.
- Words with similar meaning have similar vectors.



Machine Learning Models for Sentiment

- Logistic Regression — simple, strong baseline.
- Naive Bayes — fast, effective for text.
- Support Vector Machines (SVM) — powerful for high-dimensional data.
- Random Forest — ensemble of decision trees.
- Neural Networks — CNNs/RNNs for deeper text understanding.



"Now, that wasn't so hard, was it?"

Advantages and Limitation ML Approaches



- **High Accuracy** — captures complex patterns.
- **Scalable** — works well with big datasets.
- **Flexible** — adaptable to different domains/languages.



Advantages and Limitation ML Approaches



- **High Accuracy** — captures complex patterns.
- **Scalable** — works well with big datasets.
- **Flexible** — adaptable to different domains/languages.



- Needs lots of labeled data.
- **Complex models** require tuning and experimentation.
- **Lower interpretability** than rule-based systems.

Deep Learning Approaches

- Uses neural networks to automatically learn complex language patterns.
- Outperforms traditional machine learning on large and complex datasets.



Deep Learning Approaches



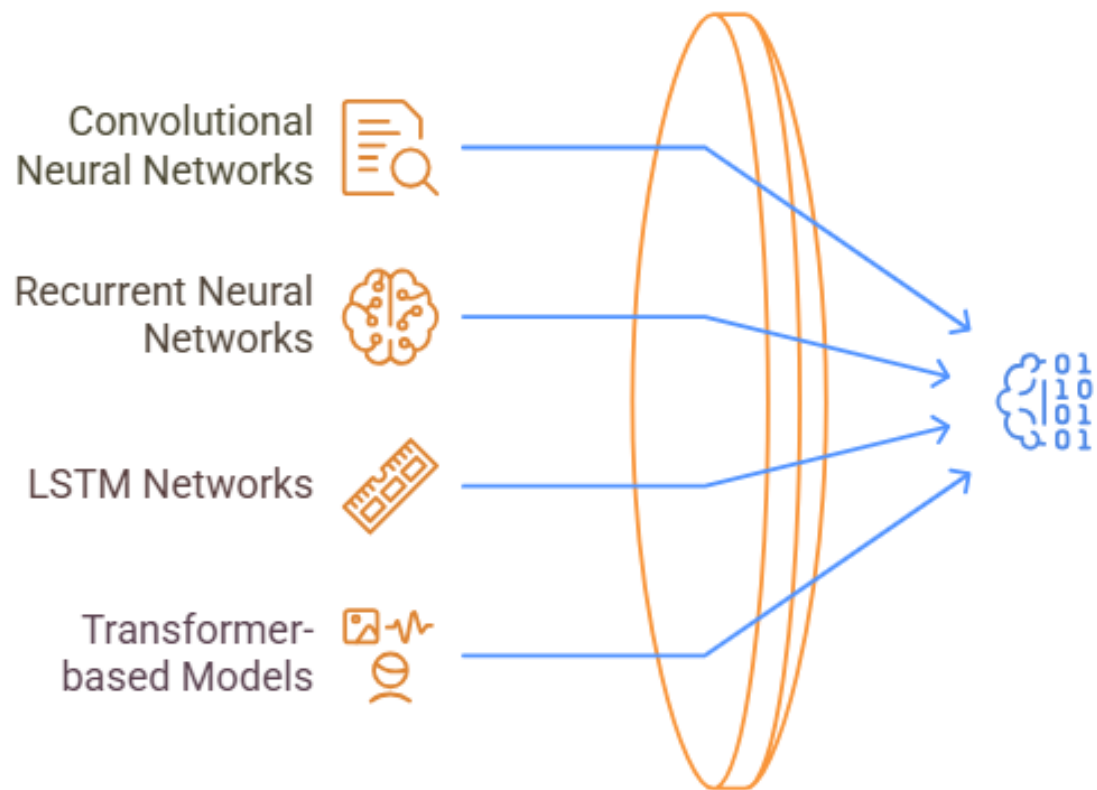
- Uses neural networks to automatically learn complex language patterns.
- Outperforms traditional machine learning on large and complex datasets.
- CNNs and RNNs can capture nuanced linguistic features for better sentiment prediction.
- Handles long-range dependencies and large vocabularies effectively.

Deep Learning Approaches



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- Outperforms traditional machine learning on large and complex datasets.
- CNNs and RNNs can capture nuanced linguistic features for better sentiment prediction.
- Handles long-range dependencies and large vocabularies effectively.
- LSTMs and Transformers maintain context over longer text sequences.
- Can be pre-trained on large corpora and fine-tuned for specific domains.
- Highly generalizable and adaptable for real-world sentiment analysis.

Deep Learning Models for Sentiment



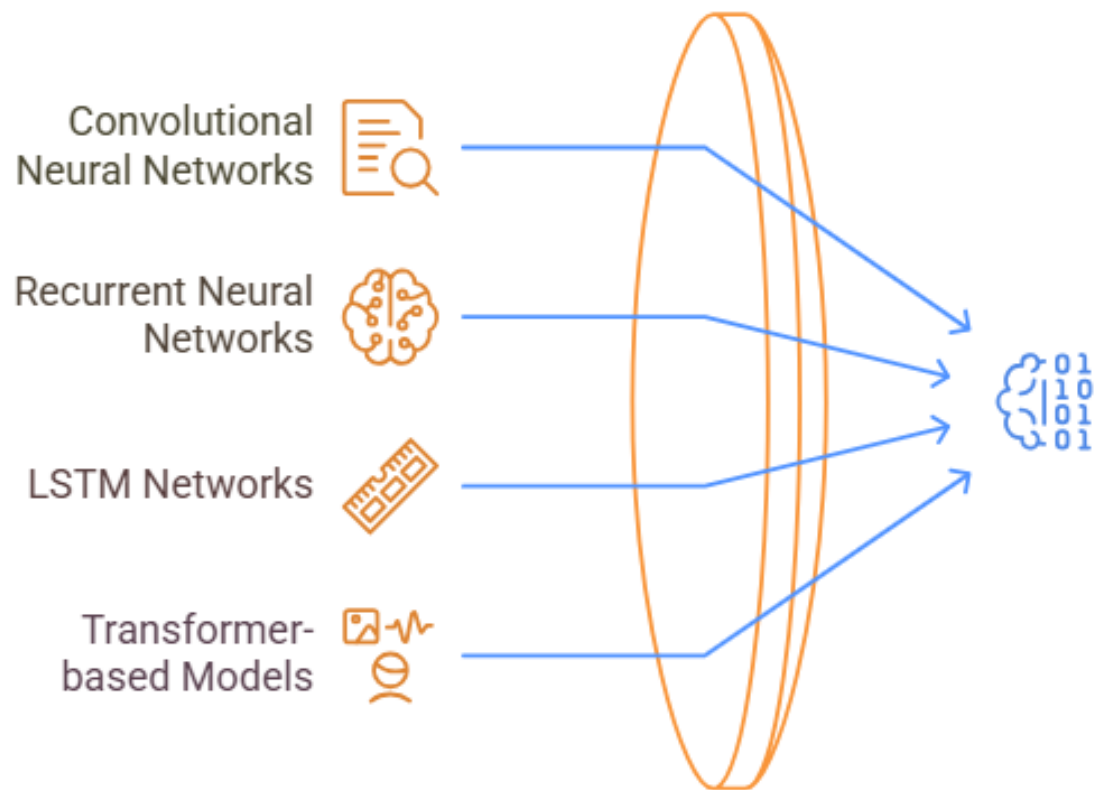
Convolutional Neural Networks (CNNs)

- Originally for images, adapted to detect local patterns in text (n-grams).
- Great for short texts and phrase-based sentiment.

Recurrent Neural Networks (RNNs)

- Designed for sequential data (process text word-by-word).
- Maintains memory of previous words.

Deep Learning Models for Sentiment



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Long Short-Term Memory (LSTM) Networks

- A special RNN that captures long-range dependencies.
- Avoids vanishing gradients → keeps important context.

Transformer-based Models (BERT)

- Uses self-attention to model relationships across the entire sentence.
- Current state-of-the-art for most NLP tasks.

Advantages and Limitation ML Approaches



- High accuracy, often state-of-the-art.
- Learns features automatically (no manual engineering needed).
- Handles complex and long text extremely well.
- Adaptable via pretraining + fine-tuning.



Advantages and Limitation DL Approaches



- High accuracy, often state-of-the-art.
- Learns features automatically (no manual engineering needed).
- Handles complex and long text extremely well.
- Adaptable via pretraining + fine-tuning.



- Requires large labeled datasets.
- Computationally expensive (GPUs/TPUs needed).
- Long training times.
- Often a black box — hard to interpret why it made a decision





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